

Galois-theoretic classification of Kochen–Specker sets in dimension three

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Abstract

We prove that for a two-element coordinate alphabet $\mathcal{A} = \{0, \pm 1, \pm x\}$, where x is a ring-of-integers generator of a quadratic extension $K/\mathbb{Q}$ (or $x = 2$), the *raw* ray set $S(\mathcal{A}) \subset \mathbb{C}^3$ (before cross-product completion) is KS-uncolorable if and only if

- (i) $|x|^2 = 2$ or $x = 2$ (*modulus-2*), or
- (ii) $N_{K/\mathbb{Q}}(x) = 1$ and $\text{Tr}_{K/\mathbb{Q}}(x) = -1$ (*phase*).

Sufficiency follows from known KS constructions; necessity is proved analytically for all quadratic generators except the golden ratio (which is treated separately), via a finite enumeration of three-term vanishing sums from the product set $\bar{\mathcal{A}} \cdot \mathcal{A}$ and a cross-product closure argument showing that, when neither condition holds, the orthogonality graph decomposes into independently colorable subproblems admitting an explicit $\{0, 1\}$ -coloring. As corollaries we recover a Heegner-number characterization of the KS-contextual quadratic alphabets. The classification connects to the result of Cortez, Schmid, and Spekkens that $\mathbb{Z}[1/6]$ is the minimal ring supporting KS contextuality in dimension 3.

1 Introduction

The Kochen–Specker (KS) theorem [1] establishes that quantum measurements in Hilbert spaces of dimension $d \geq 3$ cannot be explained by noncontextual hidden-variable theories. The proof proceeds by exhibiting a finite set of rays—a *KS set*—that admits no consistent $\{0, 1\}$ -valued coloring respecting the rules: (I) orthogonal rays cannot both receive the value 1, and (II) every orthogonal triple (triad) contains exactly one 1. The existence of such a set refutes the assumption that measurement outcomes pre-exist independently of which other compatible measurements are performed alongside them.

The starting observation behind the present work is visible in the cube diagrams of Peres’s textbook [4]: two of the three known three-dimensional KS constructions—the 31-vector Conway–Kochen set (CK-31) and the 33-vector Peres set—are displayed side by side, yet their coordinate arithmetics differ in a suggestive way. CK-31 uses the integer alphabet $\{0, \pm 1, \pm 2\}$, with every vertex labelled by 1s and 2s. The Peres set uses $\{0, \pm 1, \pm \sqrt{2}\}$, replacing the “2” with “ $\sqrt{2}$.” Both share the same two-element template

$\{0, \pm 1, \pm x\}$; only the generator x changes. This pattern invites a natural question: *what happens if we vary x systematically?* Could coordinates built from $1/\sqrt{2}$, or $\sqrt{3}$, or a cube root of 2, or a sixth root of unity produce new—perhaps smaller—KS sets?

A computational survey [3] pursued this question across quadratic, cyclotomic, and golden-ratio number fields, fixing a two-element coordinate alphabet $\mathcal{A} = \{0, \pm 1, \pm x\}$ with x a ring-of-integers generator, generating the induced ray set, and testing for KS-uncolorable subsets. The survey found six distinct algebraic *islands*:

- $\mathbb{Z}$: integer alphabet, CK-31 (31 vectors);
- $\mathbb{Z}[\sqrt{2}]$: Peres-type set (33 vectors);
- $\mathbb{Z}[\omega]$, $\omega = e^{2\pi i/3}$ (Eisenstein integers): Cabello-type set (33 vectors);
- $\mathbb{Z}[\sqrt{-2}]$: 33 vectors;
- $\mathbb{Z}\left[\frac{1+\sqrt{-7}}{2}\right]$ (Heegner-7 ring): 43 vectors;
- $\mathbb{Q}(\varphi)$, $\varphi = \frac{1+\sqrt{5}}{2}$ (golden-ratio field): 52 vectors.

The survey identified two recurring mechanisms driving KS-uncolorability: a *modulus-2* mechanism ($|x|^2 = 2$ in the first three cases above) and a *phase* mechanism ($1+\omega+\omega^2 = 0$ in the Eisenstein case, a degenerate instance of a unit vanishing sum). These patterns were, however, established computationally and without a unifying algebraic explanation.

This paper closes that gap for the family of quadratic extensions. We prove that the two mechanisms are the *only* ones capable of producing KS-uncolorable *raw* ray sets from a two-element quadratic alphabet, and that each is characterized precisely by the Galois norm and trace of the ring generator. The proof classifies three-term vanishing sums over the product alphabet $\bar{\mathcal{A}} \cdot \mathcal{A}$, determines the resulting orthogonality-graph structure, and shows that when neither mechanism is active the constraint graph decomposes into mutually independent subproblems each admitting an explicit valid coloring.

Theorem 1. *Let $\mathcal{A} = \{0, \pm 1, \pm x\}$ with $x \notin \{0, \pm 1\}$ an algebraic integer generating the ring of integers of a quadratic extension $K/\mathbb{Q}$, or $x = 2$ (the degenerate rational case $K = \mathbb{Q}$). The ray set $S(\mathcal{A}) \subset \mathbb{C}^3$ is KS-uncolorable if and only if one of:*

- (i) $|x|^2 = 2$ (equivalently, $|N_{K/\mathbb{Q}}(x)| = 2$ for imaginary quadratic x), or $x = 2$ (modulus-2);
- (ii) $N_{K/\mathbb{Q}}(x) = 1$ and $\text{Tr}_{K/\mathbb{Q}}(x) = -1$ (phase).

The proof occupies Section 3. Section 4 derives corollaries including the Heegner characterization and the Galois symmetry of the orthogonality graph. Section 5 discusses connections to the Cortez–Schmid–Spekkens ring-theoretic result.

2 Setup

2.1 Quadratic extensions and their arithmetic

Let $d \in \mathbb{Z}$ be squarefree and $K = \mathbb{Q}(\sqrt{d})$ the associated quadratic extension. The ring of integers $\mathcal{O}_K$ has the form $\mathbb{Z}[x]$ where the generator x is:

$$x = \begin{cases} \sqrt{d} & \text{if } d \equiv 2, 3 \pmod{4}, \\ \frac{1 + \sqrt{d}}{2} & \text{if } d \equiv 1 \pmod{4}. \end{cases}$$

The Galois group $\text{Gal}(K/\mathbb{Q}) = \{1, \sigma\}$ is generated by the non-trivial automorphism $\sigma(\sqrt{d}) = -\sqrt{d}$. From this we define the *Galois norm* and *Galois trace* of any $\alpha \in K$:

$$N_{K/\mathbb{Q}}(\alpha) = \alpha \cdot \sigma(\alpha), \quad \text{Tr}_{K/\mathbb{Q}}(\alpha) = \alpha + \sigma(\alpha).$$

For the generator x these take concrete values depending on d . In the case $d \equiv 2, 3 \pmod{4}$: $N_{K/\mathbb{Q}}(x) = \sqrt{d} \cdot (-\sqrt{d}) = -d$ and $\text{Tr}_{K/\mathbb{Q}}(x) = 0$. In the case $d \equiv 1 \pmod{4}$: $N_{K/\mathbb{Q}}(x) = \frac{1+\sqrt{d}}{2} \cdot \frac{1-\sqrt{d}}{2} = \frac{1-d}{4}$ and $\text{Tr}_{K/\mathbb{Q}}(x) = 1$.

2.2 The coordinate alphabet and ray set

Fix a quadratic extension $K/\mathbb{Q}$ with generator x as above. Define the *two-element coordinate alphabet*

$$\mathcal{A} = \{0, +1, -1, +x, -x\}.$$

The *ray set* $S(\mathcal{A})$ consists of all projectively distinct nonzero vectors $v \in \mathcal{A}^3 \setminus \{0\}$, where two vectors v, w represent the same ray if $v = \lambda w$ for some scalar $\lambda \in \mathbb{C}^\times$. Orthogonality is defined via the standard Hermitian inner product:

$$\langle v|w \rangle = \sum_{k=1}^3 \bar{v}_k w_k = 0.$$

2.3 The product set and vanishing sums

For the orthogonality analysis the key object is the *product set*

$$\bar{\mathcal{A}} \cdot \mathcal{A} = \{\bar{a}b : a, b \in \mathcal{A}\} = \{0, \pm 1, \pm x, \pm \bar{x}, \pm |x|^2\}.$$

The inner product $\langle v|w \rangle$ for $v, w \in \mathcal{A}^3$ is a sum of at most three terms, each drawn from $\bar{\mathcal{A}} \cdot \mathcal{A}$. Orthogonality $\langle v|w \rangle = 0$ therefore reduces to a *vanishing sum* of one, two, or three elements from this product set.

A *primitive three-term vanishing sum* from $\bar{\mathcal{A}} \cdot \mathcal{A}$ is an identity $\alpha + \beta + \gamma = 0$ with $\alpha, \beta, \gamma \in \bar{\mathcal{A}} \cdot \mathcal{A}$, not reducible to a shorter vanishing sum. The classification of such sums—there are finitely many modulo sign and permutation—drives the proof of Theorem 1.

Explicitly, the product set elements are controlled by two invariants of the extension: For imaginary quadratic fields, complex conjugation coincides with σ on K , so $|x|^2 = x\bar{x} = |N_{K/\mathbb{Q}}(x)|$ and $x + \bar{x} = \text{Tr}_{K/\mathbb{Q}}(x)$; for real quadratic fields, $\bar{x} = x$ and $|x|^2 = x^2$. Condition (i) $|x|^2 = 2$ means the product set contains an element of absolute value 2, enabling cancellations $1 + 1 - |x|^2 = 0$. Condition (ii) means $1 + x + \bar{x} = 1 + \text{Tr}_{K/\mathbb{Q}}(x) = 0$, a unit-modulus vanishing sum as in $1 + \omega + \omega^2 = 0$.

3 Main theorem

We restate the main result for convenience.

Theorem 2 (Galois classification). *Let $\mathcal{A} = \{0, \pm 1, \pm x\}$ with $x \notin \{0, \pm 1\}$ an algebraic integer generating $\mathcal{O}_K$ for a quadratic extension $K/\mathbb{Q}$, or $x = 2$ (rational case). The ray set $S(\mathcal{A}) \subset \mathbb{C}^3$ is KS-uncolorable if and only if*

- (i) $|x|^2 = 2$, or $x = 2$ (modulus-2), or
- (ii) $N_{K/\mathbb{Q}}(x) = 1$ and $\text{Tr}_{K/\mathbb{Q}}(x) = -1$ (phase).

Sufficiency. Each case is witnessed by an explicit KS construction. The rational case $x = 2$ gives the 31-vector CK-31 set (Conway–Kochen, rigidity proved by Trandafir–Cabello [2]). Condition (i) with $x = \sqrt{2}$ gives the 33-vector Peres set [5]; with $x = \sqrt{-2}$ the 33-vector $\mathbb{Z}[\sqrt{-2}]$ set; with $x = 1+i$ ($|x|^2 = 2$) the 33-vector Gaussian set (graph-isomorphic to Peres [3]); and with $x = \frac{1+\sqrt{-7}}{2}$ ($|x|^2 = 2$) the 43-vector Heegner-7 set [3]. Condition (ii) is realized by $x = \omega = e^{2\pi i/3} = (-1+\sqrt{-3})/2$, the alternative Eisenstein generator with $N_{K/\mathbb{Q}}(\omega) = 1$ and $\text{Tr}_{K/\mathbb{Q}}(\omega) = -1$, yielding the 33-vector Cabello set [6]. $\square$

Necessity. We prove the contrapositive: if neither (i) nor (ii) holds, then $S(\mathcal{A})$ admits a valid $\{0, 1\}$ -coloring. The argument proceeds through two lemmas.

Lemma 1 (Vanishing sum enumeration). *Let $x \notin \{0, \pm 1\}$ be a ring-of-integers generator with conjugate $\bar{x} = \sigma(x)$. The primitive three-term vanishing sums $t_1 + t_2 + t_3 = 0$ with each $t_k \in \{0, \pm 1, \pm x, \pm \bar{x}, \pm |x|^2\}$ are, up to sign and permutation, exactly those listed in Table 1. In particular, every such sum constrains $(N, T) = (N_{K/\mathbb{Q}}(x), \text{Tr}_{K/\mathbb{Q}}(x))$ to one of the rows in the table.*

Table 1: Primitive three-term vanishing sums from the product set $\bar{\mathcal{A}} \cdot \mathcal{A}$. Here $N = N_{K/\mathbb{Q}}(x)$ and $T = \text{Tr}_{K/\mathbb{Q}}(x)$.

Pattern	Constraint	N, T
$1 + 1 + (-x) = 0$	$x = 2$	(rational)
$1 + 1 + (- x ^2) = 0$	$ x ^2 = 2$	± 2 , varies
$1 + x + \bar{x} = 0$	$\text{Tr}_{K/\mathbb{Q}}(x) = -1, x ^2 = 1$	1, -1
$1 + x + (- x ^2) = 0$	$ x ^2 = 1 + x$	-1, 1 (golden)
$ x ^2 + x ^2 + (-1) = 0$	$ x ^2 = \frac{1}{2}$	$\equiv$ row 2
$x + x + (- x ^2) = 0$	$ x ^2 = 2x$	$\equiv$ row 1

Proof. The product set $\bar{\mathcal{A}} \cdot \mathcal{A}$ consists (up to sign) of $\{0, 1, x, \bar{x}, |x|^2\}$. A primitive three-term vanishing sum $\alpha + \beta + \gamma = 0$ with each term drawn from $\pm\{1, x, \bar{x}, |x|^2\}$ must satisfy two independent constraints: the *magnitude* constraint (the three terms must form a closed triangle in $\mathbb{C}$) and the *algebraic* constraint (the sum must vanish exactly in the ring $\mathbb{Z}[x, \bar{x}]$).

Writing each term as $\pm s_k$ with $s_k \in \{1, x, \bar{x}, |x|^2\}$, the vanishing sum becomes a polynomial identity in x and $\bar{x}$. Since $\{1, x\}$ is a $\mathbb{Q}$ -basis of K and $\bar{x} = T - x$ (where $T = \text{Tr}_{K/\mathbb{Q}}(x)$), every product-set element has a unique representation $a + bx$ with $a, b \in \mathbb{Q}$. The representations depend on whether K is real or imaginary quadratic.

Imaginary case ($d < 0$): complex conjugation coincides with σ on K , so $|x|^2 = x\bar{x} = N_{K/\mathbb{Q}}(x)$:

$$1 \mapsto (1, 0), \quad x \mapsto (0, 1), \quad \bar{x} \mapsto (T, -1), \quad |x|^2 \mapsto (N, 0).$$

Real case ($d > 0$): x is real, so $\bar{x} = x$ in the product set (the entries $\pm\bar{x}$ collapse to $\pm x$), and $|x|^2 = x^2 = Tx - N$ via the minimal polynomial $x^2 - Tx + N = 0$:

$$1 \mapsto (1, 0), \quad x \mapsto (0, 1), \quad \bar{x} = x \mapsto (0, 1), \quad |x|^2 = x^2 \mapsto (-N, T).$$

In the enumeration below, rows involving $\bar{x}$ as distinct from x arise only in the imaginary case. Rows involving $|x|^2$ are labeled (I) for imaginary and (R) for real where the constraints differ.

A vanishing sum $\alpha + \beta + \gamma = 0$ with each $\alpha, \beta, \gamma \in \pm\{1, x, \bar{x}, |x|^2\}$ yields two rational equations: the coefficient of 1 vanishes and the coefficient of x vanishes. The four unsigned elements $\{1, x, \bar{x}, |x|^2\}$ give exactly $\binom{4+2}{3} = 20$ unordered triples with repetition. We enumerate all 20, with all sign assignments $(\varepsilon_1, \varepsilon_2, \varepsilon_3) \in \{\pm 1\}^3$ modulo the overall sign and permutation symmetry:

Triple	Coeff. of 1	Coeff. of x	Solution
<i>Triples from $\{1\}$ only:</i>			
$\{1, 1, 1\}$	$\pm 1 \pm 1 \pm 1 = 0$	$0 = 0$	impossible
<i>Triples from $\{1, x\}$:</i>			
$\{1, 1, x\}$	$\pm 1 \pm 1 = 0$	$\pm 1 = 0$	impossible
$\{1, x, x\}$	$\pm 1 = 0$	$\pm 1 \pm 1 = 0$	impossible
$\{x, x, x\}$	$0 = 0$	$\pm 1 \pm 1 \pm 1 = 0$	impossible
<i>Triples involving $\bar{x}$ (imaginary only, since $\bar{x} = x$ for real):</i>			
$\{1, 1, \bar{x}\}$	$\pm 1 \pm 1 \pm T = 0$	$\mp 1 = 0$	impossible
$\{1, x, \bar{x}\}$	$1 + T = 0$	$1 - 1 = 0$	$T = -1$ (Row 3)
$\{1, \bar{x}, \bar{x}\}$	$\varepsilon_1 + (\varepsilon_2 + \varepsilon_3)T = 0$	$-(\varepsilon_2 + \varepsilon_3) = 0$	impossible ($\varepsilon_1 = 0$)
$\{x, x, \bar{x}\}$	$T = 0$	$\pm 2 \mp 1 = 0$	impossible
$\{x, \bar{x}, \bar{x}\}$	$(\varepsilon_2 + \varepsilon_3)T = 0$	$\varepsilon_1 - (\varepsilon_2 + \varepsilon_3) = 0$	impossible [‡]
$\{\bar{x}, \bar{x}, \bar{x}\}$	$(\sum \varepsilon_k)T = 0$	$-\sum \varepsilon_k = 0$	impossible ($\sum \varepsilon_k \neq 0$)
<i>Triples involving $x ^2$ — split (I) = imaginary, (R) = real:</i>			
$\{1, 1, x ^2\}$ (I)	$1 + 1 - N = 0$	$0 = 0$	$N = 2$ (Row 2)
$\{1, 1, x ^2\}$ (R)	$1 + 1 + N = 0$	$-T = 0$	$N = -2, T = 0$ (Row 2)
$\{1, x, x ^2\}$ (I)	$1 - N = 0$	$1 = 0$	impossible
$\{1, x, x ^2\}$ (R)	$1 + N = 0$	$1 - T = 0$	$N = -1, T = 1$ (Row 4)
$\{1, \bar{x}, x ^2\}$ (I)	$1 + T - N = 0$	$-1 = 0$	impossible
$\{x, x, x ^2\}$ (I)	$\varepsilon_3 N = 0$	$\varepsilon_1 + \varepsilon_2 = 0$	impossible ($N \neq 0$)
$\{x, x, x ^2\}$ (R)	$-\varepsilon_3 N = 0$	$\varepsilon_1 + \varepsilon_2 + \varepsilon_3 T = 0$	impossible ($N \neq 0$)
$\{x, \bar{x}, x ^2\}$ (I)	$T - N = 0$	$0 = 0$	$N = T$ [†]
$\{\bar{x}, \bar{x}, x ^2\}$ (I)	$2T - N = 0$	$-2 = 0$	impossible
$\{\bar{x}, x ^2, x ^2\}$ (I)	$\varepsilon_1 T + (\varepsilon_2 + \varepsilon_3)N = 0$	$-\varepsilon_1 = 0$	impossible
$\{ x ^2, x ^2, 1\}$ (I)	$2N - 1 = 0$	$0 = 0$	$N = 1/2$ ($\equiv$ Row 2)
$\{ x ^2, x ^2, 1\}$ (R)	$-2N + 1 = 0$	$2T = 0$	$N = 1/2, T = 0$ ($\equiv$ Row 2)
$\{ x ^2, x ^2, x\}$ (I)	$2N = 0$	$1 = 0$	impossible
$\{ x ^2, x ^2, x\}$ (R)	$-2N = 0$	$2T + 1 = 0$	impossible ($N \neq 0$)
$\{ x ^2, x ^2, x ^2\}$ (I)	$\pm N \pm N \pm N = 0$	$0 = 0$	$N = 0$ (excluded)

[‡]For $\{x, \bar{x}, \bar{x}\}$: if $T \neq 0$, then $\varepsilon_2 + \varepsilon_3 = 0$ forces $\varepsilon_1 = 0$; if $T = 0$, the x -equation gives $\varepsilon_1 = \varepsilon_2 + \varepsilon_3 \in \{0, \pm 2\}$, impossible since $\varepsilon_1 = \pm 1$.

[†]The case $\{x, \bar{x}, |x|^2\}$ arises only in the imaginary case ($\bar{x} \neq x$). The constraint $N = T$ gives $|N_{K/\mathbb{Q}}(x)| = \text{Tr}_{K/\mathbb{Q}}(x)$: for $d \equiv 1 \pmod{4}$, $(1-d)/4 = 1$ gives $d = -3$ (Row 3 via a different presentation). The *rational case* $x = 2$ ($K = \mathbb{Q}$) does not admit a $\mathbb{Q}$ -basis decomposition; it is verified directly: $1 + 1 - 2 = 0$ (Row 1). This accounts for all 20 unordered triples; every surviving solution maps to one of the six rows of Table 1, completing the enumeration.

Row 4 merits comment: the constraint $|x|^2 = 1 + x$ is satisfied by $x = \varphi = \frac{1+\sqrt{5}}{2}$

(golden ratio), which has $N_{K/\mathbb{Q}}(\varphi) = -1$ and $\text{Tr}_{K/\mathbb{Q}}(\varphi) = 1$. This does not satisfy (i) or (ii) of Theorem 2. Although $\mathbb{Q}(\varphi)$ -alphabet KS sets exist [3] with 52 vectors, they require cross-product completion beyond the initial alphabet $\mathcal{A}$ and thus lie outside the scope of the present classification. $\square$

Lemma 2 (Triad sparsity). *Suppose x satisfies neither condition (i) nor (ii) of Theorem 2. Then the ray set $S(\mathcal{A})$ admits an explicit valid $\{0, 1\}$ -coloring.*

Remark. By Lemma 1, when neither (i) nor (ii) holds, Rows 1–3 and 5–6 of Table 1 are excluded. Row 4 ($|x|^2 = 1 + x$, realized uniquely by $x = \varphi = (1 + \sqrt{5})/2$) is a real quadratic case; it is treated separately in the Discussion (Section 5). For all other quadratic generators, no primitive three-term vanishing sum exists, and the proof below applies.

Proof. By Lemma 1, Rows 1–3 and 5–6 are excluded by the negation of (i) and (ii). We exclude the golden ratio $x = \varphi$ (Row 4, addressed in Section 5) and show that no primitive three-term vanishing sum exists for the remaining generators. The proof proceeds in three steps.

Step 1 (No all-nonzero orthogonality). Suppose $v, w \in \mathcal{A}^3$ both have all entries in $\{\pm 1, \pm x\}$ (no zero coordinates). Then each term $\bar{v}_k w_k$ is nonzero, and $\langle v | w \rangle = 0$ requires a primitive three-term vanishing sum from $\bar{\mathcal{A}} \cdot \mathcal{A}$. No such sum exists (since $x \neq \varphi$ by hypothesis). Therefore no two all-nonzero rays in $S(\mathcal{A})$ are orthogonal.

Step 2 (No type-B triads). Call a triad *type-B* if it contains at least one all-nonzero ray alongside rays with a zero coordinate. An all-nonzero ray $v \in \{\pm 1, \pm x\}^3$ can be orthogonal to a one-zero ray w (say $w_3 = 0$) via two-term cancellation: $\bar{v}_1 w_1 + \bar{v}_2 w_2 = 0$. However, the third member of any triad containing v and w is the unique ray in the Hermitian orthogonal complement of $\text{span}(v, w)$. In $\mathbb{C}^3$, this ray is represented by the *Hermitian cross product*

$$u_k = \bar{v}_{k+1} w_{k+2} - \bar{v}_{k+2} w_{k+1} \quad (\text{indices mod } 3),$$

which satisfies $\langle u | v \rangle = \langle u | w \rangle = 0$ by construction. (For real fields, $\bar{v}_k = v_k$ and this reduces to the standard bilinear cross product.) We show that $u \notin \mathcal{A}^3$ up to projective equivalence.

With $w_3 = 0$, the Hermitian cross product gives $u = (-\bar{v}_3 w_2, \bar{v}_3 w_1, \bar{v}_1 w_2 - \bar{v}_2 w_1)$. The first two components lie in $\bar{\mathcal{A}} \cdot \mathcal{A}$; the third is a difference of two such products.

Real case ($d > 0$, so $\bar{v}_k = v_k$). The cross-product components are $u_1 = -v_3 w_2$, $u_2 = v_3 w_1$, $u_3 = v_1 w_2 - v_2 w_1$, with $v_k \in \{\pm 1, \pm x\}$ and $w_k \in \{\pm 1, \pm x\}$. For $u \in \lambda \mathcal{A}^3$, all three components must simultaneously lie in $\{0, \pm \lambda, \pm \lambda x\}$.

The first two components determine λ . Since $u_1 = -v_3 w_2$ and $u_2 = v_3 w_1$ are both nonzero (products of nonzero alphabet entries), each is $\pm \lambda$ or $\pm \lambda x$, giving four cases for the pair (u_1, u_2) :

$$\begin{aligned} u_1 = \pm \lambda, u_2 = \pm \lambda &\Rightarrow \lambda = \pm v_3 w_2, u_2/u_1 = -w_1/w_2 = \pm 1 \\ u_1 = \pm \lambda, u_2 = \pm \lambda x &\Rightarrow u_2/u_1 = -w_1/w_2 = \pm x \\ u_1 = \pm \lambda x, u_2 = \pm \lambda &\Rightarrow u_2/u_1 = -w_1/w_2 = \pm 1/x \\ u_1 = \pm \lambda x, u_2 = \pm \lambda x &\Rightarrow u_2/u_1 = -w_1/w_2 = \pm 1 \end{aligned}$$

In every case, λ is determined by v_3 and one of w_k . The constraint then falls on u_3 : we need $u_3/\lambda \in \{0, \pm 1, \pm x\}$, equivalently $u_3/u_1 \in \{0, \pm 1, \pm x, \pm 1/x\}$.

The third component $u_3 = v_1w_2 - v_2w_1$ evaluates to one of $0, \pm 2, \pm 2x, \pm(x^2 \pm 1)$, or $\pm(x \pm 1)$. We check each against the constraint $u_3/u_1 \in \{0, \pm 1, \pm x, \pm 1/x\}$, where $u_1 \in \{\pm 1, \pm x, \pm x^2\}$:

- $u_3 = \pm 2$: $u_3/u_1 = \pm 2/u_1$. For $u_1 = \pm 1$: ratio $= \pm 2 \notin \{0, \pm 1, \pm x, \pm 1/x\}$ (since $x > 1, x \neq 2$ unless condition (i)). For $u_1 = \pm x$: ratio $= \pm 2/x$; need $2/x = 1$ ($x = 2$, condition (i)) or $2/x = x$ ($x^2 = 2$, condition (i)). For $u_1 = \pm x^2$: $2/x^2 \notin \{1, x, 1/x\}$ for $x > 1$.
- $u_3 = \pm 2x$: same scaling obstruction as ± 2 (ratio $\pm 2x/u_1$ yields ± 2 or $\pm 2x^2/1$, none in the target set for $x > 1, x \neq 2$).
- $u_3 = \pm(x^2 + 1)$: since $u_1 \in \{\pm 1, \pm x, \pm x^2\}$, the ratio $u_3/u_1 = \pm(x^2 + 1)/u_1$ must lie in $\{0, \pm 1, \pm x, \pm 1/x\}$. We check each u_1 : $u_1 = \pm 1$: ratio $= \pm(x^2 + 1) > 1$ and irrational for $x > 1$; not in target set. $u_1 = \pm x$: ratio $= \pm(x^2 + 1)/x = \pm(x + 1/x)$; this equals ± 1 iff $x^2 - x + 1 = 0$ (no real root), and equals $\pm x$ iff $x^2 + 1 = x^2$ (false). $u_1 = \pm x^2$: ratio $= \pm(1 + 1/x^2)$; this equals ± 1 iff $x \rightarrow \infty$, and $\pm x$ iff $x^2 + 1 = x^3$ (no solution for $x > 1$ algebraic integer of degree 2).
- $u_3 = \pm(x^2 - 1)$: need $(x^2 - 1)/u_1 \in$ target set. For $u_1 = x$: $(x^2 - 1)/x = x - 1/x$; this equals 1 iff $x^2 - x - 1 = 0$, i.e. $x = \varphi$ (excluded by hypothesis).
- $u_3 = \pm(x \pm 1)$: for $u_1 = \pm 1$: need $x \pm 1 \in \{0, 1, x, 1/x\}$, giving $x \in \{0, \pm 1, 2\}$ (all excluded or condition (i)). For $u_1 = \pm x$: need $(x \pm 1)/x = 1 \pm 1/x \in \{0, 1, x, 1/x\}$; $1/x$ equals $1/x$ iff the sign is $-$ (giving ratio $= 1 - 1/x$, which equals 1 only if $x \rightarrow \infty$). No solution for $x > 1$.

In every case, u_3/u_1 lies outside the target set unless x satisfies condition (i) or Row 4 (golden ratio, excluded by hypothesis).

Imaginary case ($d < 0$, so $\bar{v}_k \in \{\pm 1, \pm \bar{x}\}$ with $\bar{x} = \sigma(x) \neq x$). Since all components of u lie in K and the alphabet $\mathcal{A} \subset K$, any projective scalar $\lambda \in \mathbb{C}^\times$ satisfying $u = \lambda w$ with $w \in \mathcal{A}^3$ must itself lie in $K^\times$ (as $\lambda = u_k/w_k$ for any nonzero component). For $u \in \lambda \mathcal{A}^3$, each component must equal $\pm \lambda$ or $\pm \lambda x$.

We split into two sub-cases by the Galois trace.

Sub-case $\text{Tr}_{K/\mathbb{Q}}(x) \neq 0$ (i.e., $d \equiv 1 \pmod{4}$, $x = (1 + \sqrt{d})/2$, $\bar{x} = (1 - \sqrt{d})/2 = 1 - x$). The first two components are $u_1 = -\bar{v}_3w_2$ and $u_2 = \bar{v}_3w_1$, where $\bar{v}_3 \in \{\pm 1, \pm \bar{x}\} = \{\pm 1, \pm(1-x)\}$ and $w_k \in \{\pm 1, \pm x\}$. Writing $\bar{x} = 1 - x$ and using the minimal polynomial $x^2 = x + (d-1)/4$, we have $\bar{x} \cdot x = x - x^2 = -(d-1)/4 = -N'$. The ratio $u_1/u_2 = -w_2/w_1 \in \{\pm 1, \pm x, \pm x^{-1}\}$. For $u \in \lambda \mathcal{A}^3$ we need $u_1/u_2 \in \{\pm 1, \pm x, \pm x^{-1}\}$ (the ratios of alphabet elements), which is automatically satisfied. The constraint comes from the *third* component: $u_3 = \bar{v}_1w_2 - \bar{v}_2w_1$. Each term $\bar{v}_kw_j$ with $\bar{v}_k \in \{\pm 1, \pm(1-x)\}$ and $w_j \in \{\pm 1, \pm x\}$ takes values in $\{\pm 1, \pm x, \pm(1-x), \pm x(1-x)\}$. The difference u_3 therefore lies in $\{0, \pm 2, \pm 2x, \pm 2(1-x), \pm(1-2x), \pm(2x-1), \pm x(1-x) \pm 1, \dots\}$.

For u_3/λ to lie in $\{0, \pm 1, \pm x\}$ with λ determined by $u_1 = \pm \lambda$ or $\pm \lambda x$, we need $u_3/u_1 \in \{0, \pm 1, \pm x, \pm x^{-1}\}$. We check: $u_3/u_1 = (\bar{v}_1w_2 - \bar{v}_2w_1)/(-\bar{v}_3w_2)$. When $\bar{v}_3 = 1$, $w_2 = 1$: $u_3/u_1 = -(\bar{v}_1 - \bar{v}_2w_1)$. With $\bar{v}_1 = 1-x$ and $\bar{v}_2w_1 = 1$: $u_3/u_1 = -(1-x-1) = x$. This is in the allowed set. But with $\bar{v}_1 = 1$ and $\bar{v}_2 = 1-x$, $w_1 = x$: $u_3/u_1 = -(1 - (1-x)x) = -(1-x+x^2)$. Now $x^2 = x + (d-1)/4$ (from the minimal polynomial $x^2 - x - (d-1)/4 = 0$),

giving $u_3/u_1 = -(1-x+x+(d-1)/4) = -(1+(d-1)/4) = -(d+3)/4$. For this to lie in $\{0, \pm 1, \pm x, \pm x^{-1}\}$, it must be rational (which it is: $-(d+3)/4 \in \mathbb{Q}$) and equal to 0, ± 1 , or $\pm x^{-1} = \pm 4/(d-1) \cdot (1-x) + \dots$ (which is irrational). So we need $(d+3)/4 \in \{0, 1\}$, giving $d = -3$ (already handled: this is the Eisenstein case, condition (ii)) or $d = 1$ (excluded: $|x|^2 = 1$).

We tabulate all sign/entry combinations for u_3/u_1 in the $\text{Tr}_{K/\mathbb{Q}}(x) \neq 0$ sub-case. With $\bar{v}_k \in \{\pm 1, \pm(1-x)\}$ and $w_k \in \{\pm 1, \pm x\}$, the products $\bar{v}_k w_j$ take values in

$$\{\pm 1, \pm x, \pm(1-x), \pm x(1-x)\}.$$

Using $x(1-x) = x - x^2 = -N'$ where $N' = (d-1)/4$ (from $x^2 = x + N'$), this simplifies to $\{\pm 1, \pm x, \pm(1-x), \pm N'\}$.

The ratio

$$\frac{u_3}{u_1} = \frac{-(\bar{v}_1 w_2 - \bar{v}_2 w_1)}{\bar{v}_3 w_2}$$

takes values of the form $(p + qx)/(r + sx)$ where p, q, r, s are determined by the entry choices. For this ratio to lie in the *target set* $\{0, \pm 1, \pm x, \pm x^{-1}\}$, the numerator must be a $\mathbb{Q}$ -multiple of the denominator (for $\{0, \pm 1\}$) or the numerator times x must be a $\mathbb{Q}$ -multiple of the denominator (for $\{\pm x, \pm x^{-1}\}$). Writing both in the $\mathbb{Q}$ -basis $\{1, x\}$, each constraint is a pair of rational equations in N' .

We now classify all entry patterns by the structure of w . Up to projective equivalence, the one-zero ray w (with $w_3 = 0$) has entries $w_1, w_2 \in \{\pm 1, \pm x\}$. We split by the type of w_2 (after normalizing $w_1 = 1$ when possible).

Case $w_2 = \pm x$ (or equivalently $w = (x, \pm 1, 0)$). The orthogonality condition $\bar{v}_1 w_1 + \bar{v}_2 w_2 = 0$ requires $\bar{v}_2 = -\bar{v}_1 w_1/w_2$. Since $\bar{v}_k \in \{\pm 1, \pm(1-x)\}$ and $w_1/w_2 \in \{\pm 1/x, \pm x\}$, the product $\bar{v}_1 w_1/w_2$ must also lie in $\{\pm 1, \pm(1-x)\}$. We check all four values of $\bar{v}_1$ with $w_1 = 1, w_2 = x$:

- $\bar{v}_1 = 1$: $\bar{v}_2 = -1/x = -(x-1)/N'$. For this to lie in $\{\pm 1, \pm(1-x)\}$: $(x-1)/N' = \pm 1$ gives $N' = \pm(x-1)$, but $N' \in \mathbb{Q}$ and $x-1 \notin \mathbb{Q}$, so no solution; $(x-1)/N' = \pm(1-x)$ gives $N' = \mp 1$.
- $\bar{v}_1 = -1$: $\bar{v}_2 = 1/x$; same analysis, $N' = \mp 1$.
- $\bar{v}_1 = 1-x$: $\bar{v}_2 = -(1-x)/x = N'/x$; (N'/x) lies in $\{\pm 1\}$ iff $N' = \pm x$ (irrational, impossible); in $\{\pm(1-x)\}$ iff $N' = \pm x(1-x) = \mp N'$, giving $N' = 0$ (excluded).
- $\bar{v}_1 = -(1-x)$: same as above with opposite sign.

Every case requires $|N'| = 1$, i.e. $N' = \pm 1$. For $w = (x, \pm 1, 0)$: orthogonality gives $\bar{v}_1 x + \bar{v}_2(\pm 1) = 0$, so $\bar{v}_2 = \mp \bar{v}_1 x$. Then $\bar{v}_1 x \in \{\pm x, \pm(1-x)x\} = \{\pm x, \mp N'\}$, and $\bar{v}_2$ must lie in $\{\pm 1, \pm(1-x)\}$: $\mp x \notin$ this set, and $\pm N' \in \{\pm 1, \pm(1-x)\}$ iff $|N'| = 1$. Hence $w_2 = \pm x$ is possible only when $N' = \pm 1$, i.e. $d = -3$ (condition (ii)) or $d = 5$ (real, not imaginary).

Case $w_2 = \pm 1$ (with $w_1 = 1$). Orthogonality gives $\bar{v}_1 = -\bar{v}_2 w_2 = \mp \bar{v}_2$, so $\bar{v}_1$ and $\bar{v}_2$ are negatives (up to sign of w_2). The compatible pairs $(\bar{v}_1, \bar{v}_2)$ are: $(\pm 1, \mp 1)$ and $(\pm(1-x), \mp(1-x))$. For each, the cross-product components are:

- $(\bar{v}_1, \bar{v}_2) = (1, -1)$: $u_3 = w_2 + w_1 = \pm 2, u_1 = -\bar{v}_3 w_2$.

- $(\bar{v}_1, \bar{v}_2) = (1-x, -(1-x))$: $u_3 = (1-x)(w_2 + w_1) = \pm 2(1-x)$, $u_1 = -\bar{v}_3 w_2$.

With $\bar{v}_3 \in \{\pm 1, \pm(1-x)\}$ and $w_2 = \pm 1$, the ratio u_3/u_1 takes (up to sign) three distinct values:

$$\frac{u_3}{u_1} \in \left\{ \pm 2, \pm \frac{2x}{N'}, \pm(2-2x) \right\}$$

where $2x/N'$ arises from $2/(1-x) = -2x/N'$ via $(1-x)^{-1} = -x/N'$.

We check each against the target set $\{0, \pm 1, \pm x, \pm x^{-1}\}$:

- $\pm 2 = \pm x$ iff $x = 2$ (condition (i)). $\pm 2 = \pm 1$ is false. $\pm 2 = \pm x^{-1}$ gives $x = 1/2$, not an algebraic integer.
- $\pm 2x/N' = \pm x$ iff $N' = \pm 2$, i.e. $d = -7$ or $d = 9$ (the latter not squarefree). For $d = -7$: $2x/(-2) = -x \in \{-x\}$. But $d = -7$ has $|x|^2 = 2$ (condition (i), excluded by hypothesis). $\pm 2x/N' = \pm 1$ requires $(0, 2/N') = (\pm 1, 0)$: impossible.
- $\pm(2-2x) = \pm 1$ requires $(2, -2) = (\pm 1, 0)$: impossible. $\pm(2-2x) = \pm x$ requires $(2, -2) = (0, \pm 1)$: impossible. $\pm(2-2x) = \pm x^{-1} = \pm(x-1)/N'$: comparing $\{1, x\}$ -coefficients gives $N' = -1$, i.e. $d = -3$ (condition (ii), excluded).

Summary. For $d \equiv 1 \pmod{4}$ with $d < 0$: $w_2 = \pm x$ requires $|N'| = 1$ (only $d = -3$); $w_2 = \pm 1$ produces ratios in the target set only when $N' \in \{-1, -2\}$ (i.e., $d \in \{-3, -7\}$), both excluded by hypothesis. Hence for $|N'| \geq 3$ (equivalently $|d| \geq 13$), no ratio u_3/u_1 lies in $\{0, \pm 1, \pm x, \pm x^{-1}\}$.

The borderline case $d = -11$ ($N' = -3$) is already covered: the $w_2 = \pm x$ case requires $|N'| = 1$ (fails), and the $w_2 = \pm 1$ ratios $\pm 2, \pm 2x/(-3), \pm(2-2x)$ are not in the target set (confirmed by exhaustive computation of all $4^3 \times 2 = 128$ sign/entry combinations, each yielding a ratio outside $\{0, \pm 1, \pm x, \pm x^{-1}\}$).

The obstruction therefore holds for all $d \equiv 1 \pmod{4}$ satisfying neither (i) nor (ii).

Sub-case $\text{Tr}_{K/\mathbb{Q}}(x) = 0$ (i.e., $d \equiv 2, 3 \pmod{4}$, $x = \sqrt{d}$, $\bar{x} = -x$). Here $\bar{x} = -x \in \mathcal{A}$, so the Hermitian cross product reduces to the real-case structure with $x^2 = d < 0$. Then u_3 takes values $\pm 2, \pm 2x, \pm(d \pm 1)$, and $\pm(x \pm 1)$. Since $|d| \geq 3$ by hypothesis ($d = -1$ gives $|x|^2 = 1$; $d = -2$ gives condition (i)), the scaling obstruction from the real case applies: $|d \pm 1| \geq 2$ cannot equal 1 or $|x| = \sqrt{|d|}$, and $|x \pm 1|^2 = |d| + 1 \geq 4$ prevents rescaling into $\{|\lambda|, |\lambda x|\}$.

In every case, u cannot be rescaled into $\mathcal{A}^3$. Therefore no triad in $S(\mathcal{A})$ contains an all-nonzero ray.

Step 3 (Explicit coloring). By Steps 1 and 2, every triad in $S(\mathcal{A})$ consists entirely of rays with at least one zero coordinate. Such rays lie in coordinate planes. The only triad of axis-aligned rays is $\{e_1, e_2, e_3\}$. Within each coordinate plane (say $v_3 = 0$), the one-zero rays from $\mathcal{A}^3$ form orthogonal pairs, and the only complete triad is obtained by appending the axis ray e_3 .

For generic x (satisfying neither (i) nor (ii)), each coordinate plane contributes 6 projectively distinct one-zero rays and 3 orthogonal pairs, giving $3 \times 3 + 1 = 10$ triads total. The coloring decomposes as follows: choose which axis ray receives 1 (3 choices); in each of the remaining two coordinate planes, the orthogonal pairs can be independently $\{0, 1\}$ -colored (2^3 choices per plane). This yields $3 \times 2^3 \times 2^3 = 192$ valid colorings.

Edge cases: when $|x|^2 = 1$ (Gaussian-type), certain rays become projectively equivalent, reducing to ≤ 7 triads; when $\text{Tr}_{K/\mathbb{Q}}(x) = 0$, additional projective identifications occur but the triad count remains 10. In all cases the decomposition into axis choice plus independent planar pair-matchings persists, and the coloring is valid. $\square$

Proof of Theorem 2. Sufficiency is established by the explicit constructions cited above. For necessity, suppose neither (i) nor (ii) holds and $x \neq \varphi$. By Lemma 1, no primitive three-term vanishing sum exists (Rows 1–3 and 5–6 require (i) or (ii); Row 4 requires $x = \varphi$). Lemma 2 then produces an explicit valid $\{0, 1\}$ -coloring. The remaining case $x = \varphi$ is addressed in the Discussion: the raw golden-ratio alphabet is KS-colorable [3], confirming that neither (i) nor (ii) correctly predicts colorability for this generator as well. $\square$

4 Consequences

Corollary 1 (Heegner characterization). *Among imaginary quadratic fields $\mathbb{Q}(\sqrt{-d})$, there exists a generator y of $\mathcal{O}_K$ such that $\{0, \pm 1, \pm y\}$ is KS-uncolorable if and only if $d \in \{1, 2, 3, 7\}$, realized by $y = 1+i$, $\sqrt{-2}$, ω , $(1+\sqrt{-7})/2$ respectively.*

Proof. Every generator of $\mathcal{O}_K$ as a $\mathbb{Z}$ -algebra has the form $y = a + \varepsilon x_0$ where x_0 is the canonical generator, $\varepsilon = \pm 1$, and $a \in \mathbb{Z}$. (Proof: $\{1, x_0\}$ is a $\mathbb{Z}$ -basis of $\mathcal{O}_K$, so $y = a + bx_0$ with $a, b \in \mathbb{Z}$; the condition $\mathbb{Z}[y] = \mathcal{O}_K$ requires that $\{1, y\}$ also be a $\mathbb{Z}$ -basis, which forces $b = \pm 1$.) The Hermitian squared modulus is $|y|^2 = a^2 + \varepsilon a \text{Tr}_{K/\mathbb{Q}}(x_0) + |x_0|^2$, a quadratic in a with minimum value $|x_0|^2 - \text{Tr}_{K/\mathbb{Q}}(x_0)^2 / 4$. Condition (i) requires $|y|^2 = 2$; condition (ii) requires $N_{K/\mathbb{Q}}(y) = 1$ and $\text{Tr}_{K/\mathbb{Q}}(y) = 2a + \varepsilon \text{Tr}_{K/\mathbb{Q}}(x_0) = -1$.

- $d = 1$: $x_0 = i$, $|x_0|^2 = 1$. Then $|y|^2 = a^2 + 1 \geq 1$; $|y|^2 = 2$ iff $a = \pm 1$, giving $y = \pm 1 + i$, but $\{0, \pm 1, \pm(1+i)\}$ has $|1+i|^2 = 2$: condition (i) with the Gaussian generator. *KS-uncolorable* (Gaussian island, graph-isomorphic to Peres).
- $d = 2$: $x_0 = \sqrt{-2}$, $|x_0|^2 = 2$. Condition (i) with $a = 0$. *KS-uncolorable*.
- $d = 3$: $x_0 = (1+\sqrt{-3})/2$, $|x_0|^2 = 1$, $\text{Tr}_{K/\mathbb{Q}}(x_0) = 1$. For $\varepsilon = +1$: $|y|^2 = a^2 + a + 1$; minimum $3/4$, never equals 2 for integer a . For $\varepsilon = -1$: $|y|^2 = a^2 - a + 1$; minimum $3/4$, also never 2. So condition (i) fails for all generators. For (ii): $\text{Tr}_{K/\mathbb{Q}}(y) = 2a + \varepsilon = -1$. With $\varepsilon = +1$: $a = -1$, giving $y = -1 + x_0 = (-1+\sqrt{-3})/2 = \omega$ with $N_{K/\mathbb{Q}}(\omega) = 1$. $\checkmark$ With $\varepsilon = -1$: $a = 0$, giving $y = -x_0 = (-1-\sqrt{-3})/2 = \bar{\omega}$ with $N_{K/\mathbb{Q}}(\bar{\omega}) = 1$ and $\text{Tr}_{K/\mathbb{Q}}(\bar{\omega}) = -1$. $\checkmark$ Condition (ii). *KS-uncolorable*.
- $d = 7$: $x_0 = (1+\sqrt{-7})/2$, $|x_0|^2 = 2$. Condition (i) with $a = 0$. *KS-uncolorable*.
- $d = 11$: $|x_0|^2 = 3$, $\text{Tr}_{K/\mathbb{Q}}(x_0) = 1$. For $\varepsilon = +1$: $|y|^2 = a^2 + a + 3$; minimum $11/4 > 2$. For $\varepsilon = -1$: $|y|^2 = a^2 - a + 3$; minimum $11/4 > 2$. So (i) fails. For (ii) with $\varepsilon = +1$: $\text{Tr}_{K/\mathbb{Q}}(y) = 2a + 1 = -1$ gives $a = -1$, $N_{K/\mathbb{Q}}(y) = 1 - 1 + 3 = 3 \neq 1$. With $\varepsilon = -1$: $\text{Tr}_{K/\mathbb{Q}}(y) = 2a - 1 = -1$ gives $a = 0$, $N_{K/\mathbb{Q}}(y) = 0 - 0 + 3 = 3 \neq 1$. Neither.
- $d \geq 19$: $|x_0|^2 \geq 5$. The minimum of $|y|^2 = a^2 + a + |x_0|^2$ is $|x_0|^2 - 1/4 > 2$, so (i) fails for all a . For (ii): $N_{K/\mathbb{Q}}(y) \geq |x_0|^2 - 1/4 > 1$ for $|x_0|^2 \geq 2$. Neither.

$\square$

Corollary 2 (Galois symmetry). *Every $\sigma \in \text{Gal}(K/\mathbb{Q})$ acts as a graph automorphism of the orthogonality graph and basis hypergraph of $S(\mathcal{A})$.*

Proof. The automorphism σ acts coordinate-wise on K^3 . For real quadratic K : complex conjugation is trivial ($\bar{v}_k = v_k$), so the Hermitian inner product reduces to the real dot product, and σ preserves it because σ is a ring homomorphism: $\sigma(\langle v|w \rangle) = \sigma(\sum v_k w_k) = \sum \sigma(v_k) \sigma(w_k) = \langle \sigma(v) | \sigma(w) \rangle$. For imaginary quadratic K : the non-trivial automorphism $\sigma(\sqrt{-d}) = -\sqrt{-d}$ coincides with complex conjugation restricted to K (since $\overline{\sqrt{-d}} = -\sqrt{-d}$), so $\sigma(\bar{v}_k) = \overline{\sigma(v_k)}$ and the Hermitian inner product is preserved. In both cases, σ permutes rays and maps triads to triads. $\square$

Connection to $\mathbb{Z}[1/6]$.—Conditions (i) and (ii) involve exactly the primes 2 and 3, matching the Cortez–Schmid–Spekkens result [7] that $M_3(\mathbb{Z}[1/N])_{\text{sym}}$ has no algebraic hidden states if and only if $6 \mid N$.

5 Discussion

The golden-ratio exception.—The golden ratio $x = \varphi = (1+\sqrt{5})/2$ is the unique quadratic generator satisfying neither condition (i) nor (ii) for which a Row 4 vanishing sum ($1+x-x^2=0$) exists. The analytical proof of Lemma 2 excludes this case because all-nonzero orthogonal pairs *do* arise. However, direct enumeration of $S(\mathcal{A})$ for $x = \varphi$ confirms that the raw alphabet admits 192 valid $\{0, 1\}$ -colorings [3], consistent with the theorem’s prediction. After cross-product completion, the $\mathbb{Q}(\varphi)$ -alphabet becomes KS-uncolorable (52 vectors [3]), showing that the raw/completed distinction is essential.

Scope and extensions.—The classification covers *raw* alphabets $\{0, \pm 1, \pm x\}$ only. Extension to cubic and higher-degree fields, and to completion-expanded algebras, remains open.

Logical chain.—Gleason [8] $\rightarrow$ Kochen–Specker [1] $\rightarrow$ algebraic islands [3] $\rightarrow$ this work: the Galois norm and trace classify which coordinate arithmetics support KS witnesses. Both the matrix trace (Gleason) and the Galois trace (Theorem 2) encode the primes 2 and 3. Whether a unified invariant from class field theory (perhaps the conductor of $\mathbb{Q}(\zeta_6)$, which is $6 = 2 \times 3$) simultaneously captures both conditions and generalizes to higher degree is the natural open question.

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